

Heat — Lesson 3

ICSE Class 7 Physics · 10 MCQs · 1 mark each · Name: _____ Date: _____

Syllabus: Three modes of heat transfer (conduction, convection, radiation), conductors & insulators, black/white surfaces, the thermos flask.

Instructions: Each question has one correct option. Tick or circle the letter of the best answer. Attempt all questions.

1. Heat passes along a metal spoon dipped in hot tea mainly by:

- | | |
|----------------|---------------|
| A. radiation | B. conduction |
| C. evaporation | D. convection |

2. Which of the following is the best conductor of heat?

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|-----------|------------|
| A. copper | B. plastic |
| C. wood | D. glass |

3. Woollen clothes keep us warm in winter because wool:

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|---|------------------|
| A. is a good conductor | B. reflects heat |
| C. traps air and is a poor conductor, so it reduces heat loss | D. produces heat |

4. Convection can take place in:

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|----------------------|----------------|
| A. liquids and gases | B. solids only |
| C. a vacuum | D. metals |

5. Ventilators are placed near the ceiling of a room so that:

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|--------------------|--|
| A. light enters | B. warm, used air which rises can escape |
| C. cold air enters | D. the room looks bigger |

6. We feel the warmth of a fire instantly mainly because of:

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|---------------------------|--------------------|
| A. conduction through air | B. convection only |
| C. sound | D. radiation |

7. The cooling pipes at the back of a refrigerator are painted black so that they:

- | | |
|--|------------------------------|
| A. radiate (give out) heat efficiently | B. look nice |
| C. conduct electricity | D. absorb heat from the room |

8. Houses in hot countries are often painted white because white surfaces:

- | | |
|-----------------|---|
| A. radiate heat | B. reflect most of the heat and keep the house cool |
| C. absorb heat | D. conduct heat |

9. The stopper of a thermos flask is made of cork or plastic to reduce heat loss by:

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|---------------|--|
| A. reflection | B. absorption |
| C. radiation | D. conduction and convection through the mouth |

10. On a sunny day a person in black clothes feels hotter than one in white clothes because black:

- | | |
|--------------------------------|------------------|
| A. reflects heat | B. conducts less |
| C. absorbs more heat radiation | D. radiates more |

Answer Key

1. B

2. A

3. C

4. A

5. B

6. D

7. A

8. B

9. D

10. C